

ADITHYA BHAT

Visa Research, Visa Inc., CA

◇ Email - dth.bht@gmail.com ◇ GitHub - <https://github.com/libdist-rs>

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SUMMARY

Distributed systems engineer and researcher with 8+ years of experience designing, building, and evaluating fault-tolerant, high-performance distributed systems. Published 13 peer-reviewed papers at top-tier venues (CCS, IEEE S&P, NDSS, FC) and hold 3 US patents. Deep expertise in consensus protocols, state machine replication, data synchronization, and distributed data infrastructure. Proficient in Go, Rust, C/C++, and Python.

TECHNICAL SKILLS

Languages	Go, Rust, C/C++, Python, Java
Distributed Systems	Consensus protocols (BFT/CFT), State Machine Replication, Sharding, Replication, Fault-tolerant systems, Async & Sync protocols
Data Infrastructure	Key-value stores, Distributed databases, Data synchronization, Cache-coherence, Protocol optimization, Multi-threading & concurrency
Tools & Frameworks	libp2p, gRPC, Protobuf, Ethereum, Concord-BFT, Git, Linux, Docker

WORK EXPERIENCE

Visa Research, Foster City (Aug 2023 - Present)
Staff Research Scientist

- Designed and built production-grade distributed systems for payments infrastructure, including an efficient transaction replay system (Ira) and a flexible payment protocol (FPS) for Visa's global network.
- Developed FastSync, a high-throughput blockchain synchronization protocol that reduces sync time from hours to minutes, enabling rapid state recovery across distributed nodes. Filed US patent.
- Architected a partially synchronous sharding protocol and implemented Instachain, a scalable blockchain execution system that partitions workload across shards for parallel processing.
- Led research on energy-efficient consensus and homomorphic encryption applied to smart contracts (SoK accepted at DeFi 2026), bridging cutting-edge cryptography with practical system design.
- Authored 5 technical reports and patent applications; presented at RWC 2025, Dagstuhl, and Espresso Systems.

VMware Research, Remote (May 2021 - Aug 2021)
Research Intern

- Built a prototype anonymous token system on top of Concord-BFT, VMware's production Byzantine fault-tolerant distributed platform.
- Designed Quick-Pay, a one-round-trip low-latency payment protocol, and extended it into a two-phase lock-free sharding solution—eliminating coordination bottlenecks in distributed transaction processing.

Purdue University, West Lafayette (Aug 2018 - Aug 2023)
Graduate Research Assistant – Department of Computer Science

- Designed and implemented multiple fault-tolerant consensus protocols (Apollo, Artemis, Leto, OptRand, RandPiper) handling Byzantine faults in synchronous and asynchronous network settings.

- Built [libnet-rs](#): a high-performance networking library in Rust for state machine replication protocols, enabling rapid prototyping and benchmarking of distributed systems.
- Built [libstorage-rs](#) (Rust) and [libe2c-cpp](#) (C++): pluggable storage and consensus frameworks for simulating and evaluating distributed protocols at scale.
- Developed mathematical models for protocol optimization, improving energy efficiency of state machine replication (published at Middleware 2023). Directly applicable to reducing infrastructure cost at scale.
- Published 10 papers at top-tier venues including CCS (x4), IEEE S\&P, NDSS, and Financial Cryptography. Served on program committees for CCS, FC, and USENIX Security.

EDUCATION

Ph.D., Computer Science – Purdue University, West Lafayette (2018-2023)
 Advisor: Aniket Kate. Focus: Byzantine fault-tolerant distributed systems and applied cryptography.

B.Tech., Information Technology – NIT Karnataka, Surathkal (2014-2018)

SELECTED PUBLICATIONS & PATENTS

13 peer-reviewed publications at CCS (x4), IEEE S&P, DSN, ICDCS, NDSS, Middleware, FC, and Journal of Cryptology. 3 US patents. Selected highlights:

- **EESMR – Energy Efficient State Machine Replication** (Middleware 2023). Optimization models to reduce resource consumption in replicated systems.
- **OptRand – Optimistically Responsive Distributed Randomness** (NDSS 2023). Reconfigurable distributed protocol with optimistic fast-path performance.
- **The Unique Chain Rule and Its Applications** (FC 2023). Novel consistency primitive for partially synchronous distributed systems.
- **Synchronous DKG without Broadcasts** (Journal of Cryptology 2024). Efficient distributed coordination without expensive broadcast primitives.
- **Fast Sync Blockchain System and Method** – US Patent WO2024026321A1. Efficient state synchronization protocol for distributed nodes.

OPEN SOURCE & SOFTWARE ARTIFACTS

- [libnet-rs](#) (Rust) – Core networking library for distributed systems. Powers multiple BFT protocol implementations including Apollo and Sync HotStuff.
- [libstorage-rs](#) (Rust) – Pluggable storage backend library for distributed protocol nodes.
- [libmempool-rs](#) (Rust) – Transaction mempool for buffering, ordering, and managing pending operations in distributed systems.
- [libcrypto-rs](#) (Rust) – Cryptographic primitives library with traits and wrappers used across the libdist-rs ecosystem.
- [libe2c-cpp](#) (C++) – Implementation of energy-efficient consensus protocol (E2C).